(a)

Input image



H × W × 3

ResNet-18 Backbone

S5 /32 C=512

1×1 AIFI 1×1

F5 C=256

S4 /16 C=256

S3 /8 C=128

Original CCFF (replaced)

FDPN adaptation

Raw S5 skip C=512

Focus Diffuse Refocus

O5 /32 C=256

O4 /16 C=576

O3 /8 C=256

Input projections d=256

Multi-scale memory

Query selection Q=300

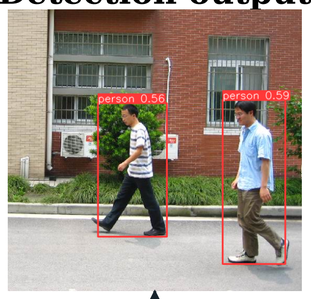
Object queries

RT-DETR decoder 3 layers d=256

Class head Q × K, K=1

Box head Q × 4

Detection output



Feature map | Attention / encoder | FDPN adaptation | Query selection | Decoder | Prediction head

(b) FDPN: focus → diffuse → refocus

Stage I: focus and redistribute

F5 /32 C=256

S4 /16 C=256

S3 /8 C=128

FocusFeature Stage 1 / L11 /16 · C=384

S5 / L7 C=512

Conv 3×3 s=2 D5 / L12

Concat + RepC3

V5 /32 C=256

V4 /16 C=384

Upsample ×2 D3 / L15

Concat + RepC3

V3 /8 C=256

S3 skip

Stage II: refocus and fuse

FocusFeature Stage 2 / L18 /16 · C=576

D5 L12

V5 L14

Conv 3×3 s=2 L19

Concat + RepC3

O5 /32 C=256

O4 /16 C=576

D3 L15

V3 L17

Upsample ×2 L22

Concat + RepC3

O3 /8 C=256

Labelled skip ports reuse the indicated upstream tensors.

FocusFeature: stage-1 detail

S3 /8 C=128

S4 /16 C=256

F5 /32 C=256

ADown /2

1×1 Conv

Upsample ×2 + 1×1 Conv

Concat → A at stride 16

DW 5×5

DW 7×7

DW 9×9

DW 11×11

Identity

Σ

1×1 Conv + BN + SiLU

+

Output

Φ(A)

C_a=384 (stage 1); 576 (stage 2)